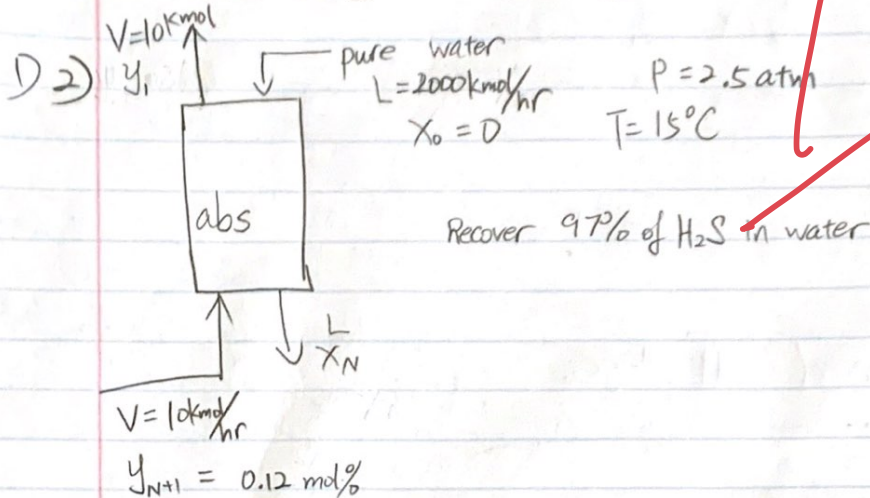


CHE 313 HW 8

Fu Wu



a) Find y_1 & X_N

$$y_1 = y_{N+1} (1 - 0.97) = 0.12\% (0.03) = 0.0036 \text{ mol\%}$$

$$\begin{aligned}
 L \cdot X_0 + V \cdot y_{N+1} &= V \cdot y_1 + L \cdot X_N \\
 10 \text{ kmol/hr} \cdot 0 &+ 10 \cdot 0.12 = 10 \cdot 0.0036 + 2000 \cdot X_N \\
 1.2 &= 0.036 + 2000 X_N \\
 1.2 - 0.036 &= 2000 X_N \\
 X_N &= \frac{1.164}{2000} = 0.000582
 \end{aligned}$$

b) # of stages = ?

at $T = 15^\circ\text{C}$ H for $\text{H}_2\text{S} = 423$

$$\begin{aligned}
 y &= \frac{H_i}{P_{\text{tot}}} X_i \\
 0.12 &= \frac{423}{2.5} X \\
 X_N^* &= 7.0922 \times 10^{-4}
 \end{aligned}$$

equilibrium line $\Rightarrow y = 169.2 X$

equilibrium line cross $(0,0)$ $(7.0922 \times 10^{-4}, 0.12)$

$$\frac{L}{V} = \frac{2000}{10} = 200$$

op line: $y = \left(\frac{L}{V}\right)x + \left(y_1 - \left(\frac{L}{V}\right)x_0\right)$ b/c $x_0 = \text{pure water}$
no H_2S

$$y = 200x + 0.0036$$

$$N \sim 10.5$$

you may need 11 stages.

$$c) \frac{L}{V}_{\min} = \frac{y_{N+1} - y_1}{x_N^* - x_0} = \frac{0.12 - 0.0036}{0.00070922} = 164.1239672$$

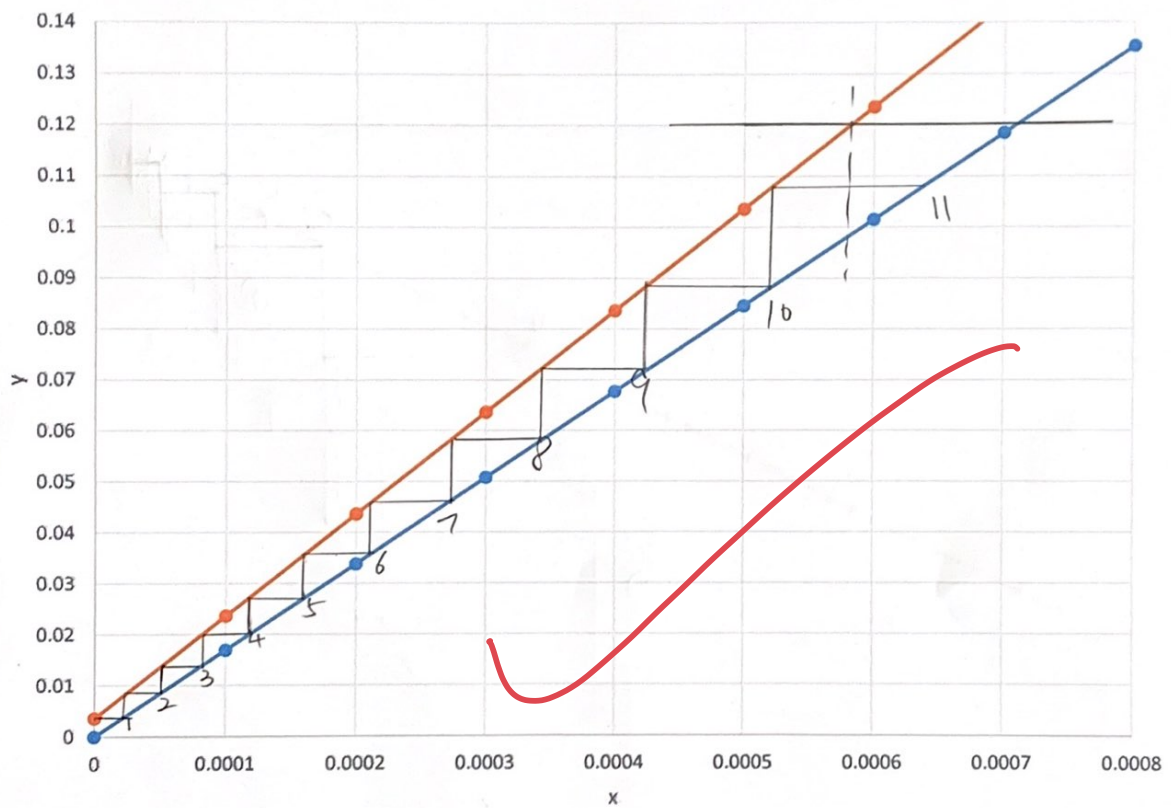
$$\frac{L}{V} = m \left(\frac{L}{V}\right)_{\min}$$

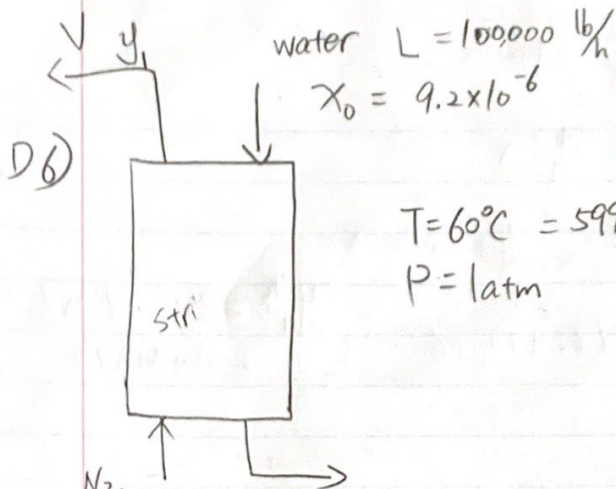
$$m = \frac{L}{V} \div \left(\frac{L}{V}\right)_{\min}$$

$$m = 200 \div 164.1239672 = 1.2186$$

- d) a) because L is way higher than V , the liquid comes out will be so dilute
b) use another solvent which requires less amount of solvent.

D2





MT method

at $T = ^\circ\text{C}$

$$\frac{1 \text{ mol}}{22.4 \text{ L}} \cdot \frac{1 \text{ L}}{0.0353147 \text{ ft}^3} = \frac{1 \text{ mol}}{0.79104928 \text{ ft}^3} \cdot \frac{2.2 \times 10^{-3} \text{ lbmol}}{1 \text{ mol}} = \frac{2.2 \times 10^{-3} \text{ lbmol}}{0.79105 \text{ ft}^3}$$

$$\frac{1 \text{ lbmol}}{359.5678545 \text{ ft}^3} \Rightarrow 1 \text{ lbmol} = 359.5678545 \text{ ft}^3$$

at 60°C

$$359.5678545 \left(\frac{333.15}{273.15} \right) = 438.55 \text{ ft}^3/\text{lbmol}$$

$$\dot{V}_{N_2} = 2500 \text{ ft}^3/\text{hr} / 438.55 \text{ ft}^3/\text{lbmol} = 5.7006 \text{ lbmol/hr} = V$$

$$\dot{m}_{H_2O} = 100,000 \text{ lb/hr} \div 18.02 \text{ lb/mol} = 5549.389567 \text{ mol/hr} = L$$

$$L \cdot X_0 + V \cdot y_{N+1} = L \cdot X_N + V \cdot y_1$$

$$y_1 = \frac{L \cdot X_0 - L \cdot X_N}{V} = \frac{5549.389567 \cdot 9.2 \times 10^{-6} - 5549.389567 \cdot 2.0 \times 10^{-7}}{5.7006}$$

$$y_1 = 0.0087612718$$

H constant for CO_2 at 65°C = 3410

$$y_i = \frac{H}{P_{\text{tot}}} x_i \quad y = \frac{3410}{1} x$$

$\Rightarrow$ equilibrium line: $y = 3410 x$

$$\frac{L}{V} = \frac{5549.389567}{5.7006} = 973.474646$$

$$y_1^* = 3410 \cdot 9.2 \times 10^{-6} \\ = 0.031372$$

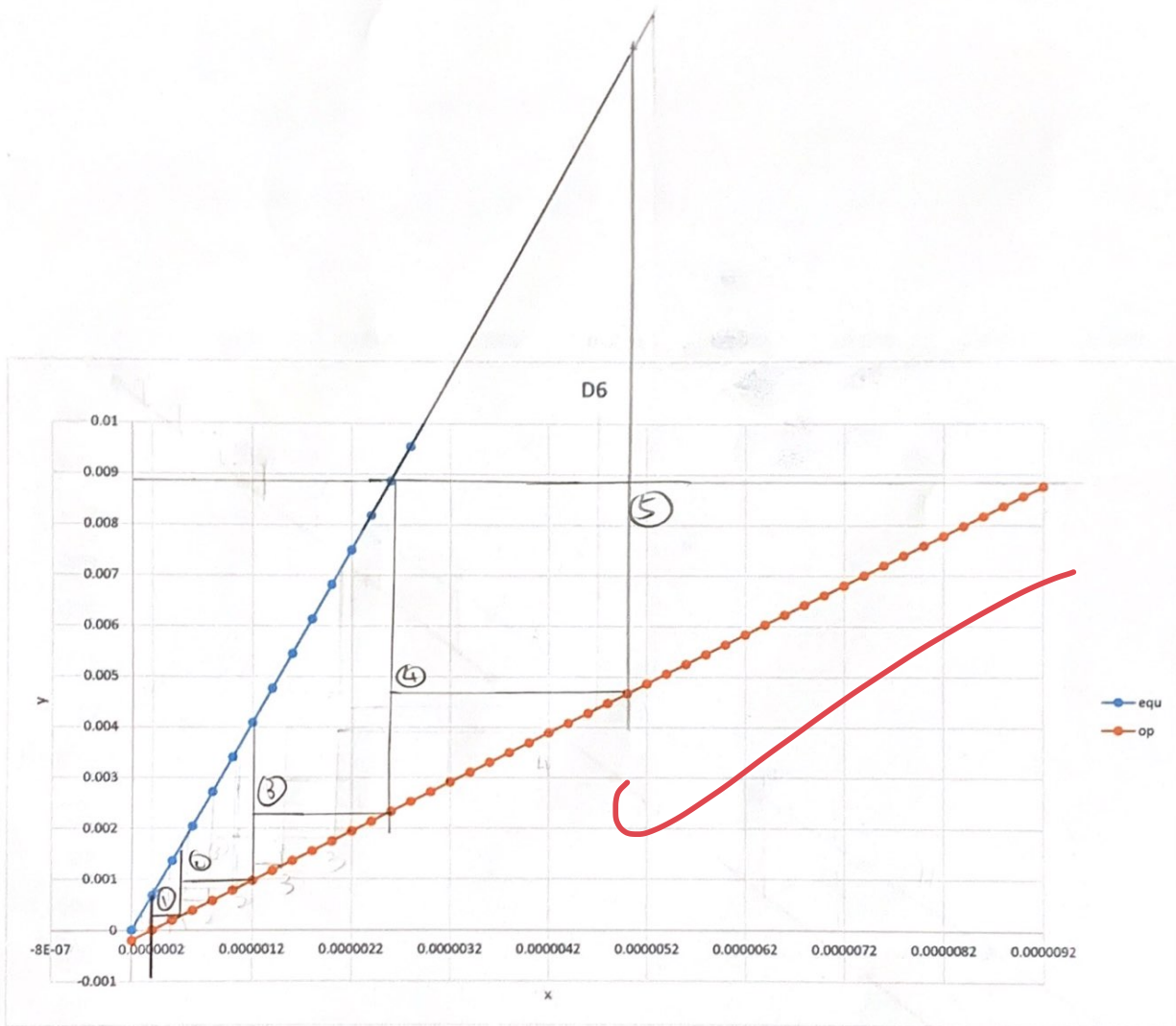
op line

$$y = \left(\frac{L}{V}\right)x + \left(y_1 - \left(\frac{L}{V}\right)x_0\right)$$

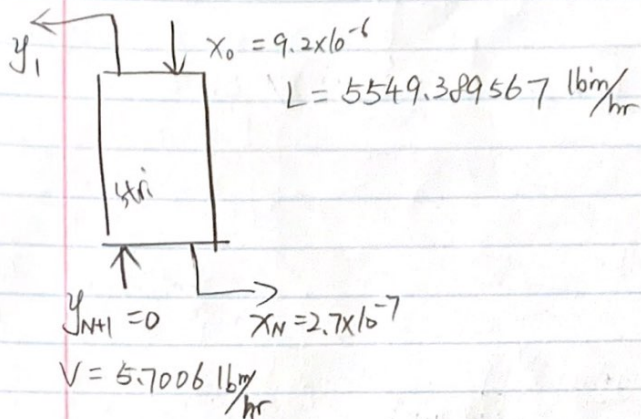
$$y = 973.474646x + 0.0087612718 + 973.474646 \cdot 9.2 \times 10^{-6}$$

$$y = 973.474646x - 1.947 \times 10^{-4}$$

$$y = 973.474646x - 0.0001947$$



Kremser method:



$$\frac{V}{L} = \frac{5.7006}{5549.389567} = 0.0010272481$$

$$\frac{L}{V} = 973.474646$$

$$y = \frac{H}{P_{\text{tot}}} x = \frac{3410}{1} x = 3410x$$

$$y = 3410x \quad m = 3410 \quad y_1 = 0.0087612718$$

$$y_1^* = 3410 X_0 = 3410 \cdot 9.2 \times 10^{-6} = 0.031372$$

$$E_{MV} = 0.4$$

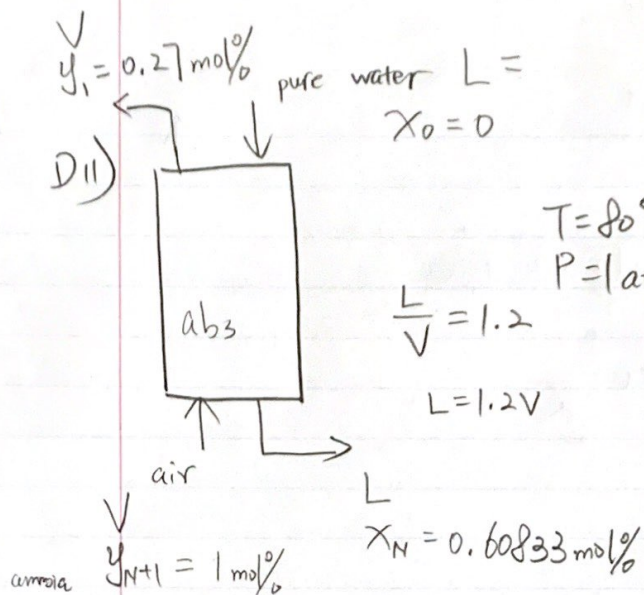
$$N = \frac{\ln \left[\left(1 - \frac{mV}{L} \right) \left(\frac{y_{N+1} - y_1^*}{y_1 - y_1^*} \right) + \frac{mV}{L} \right]}{\ln \left[1 + E_{MV} \left(\frac{mV}{L} - 1 \right) \right]}$$

$$\frac{mV}{L} = 3.502916089$$

$$\frac{L}{mV} = 0.2854764358$$

$$N = \frac{\ln \left[(1 - 3.502916089) \left(\frac{0 - 0.031372}{0.0087612718 - 0.031372} \right) + 3.502916089 \right]}{\ln [1 + 0.4 (3.502916089 - 1)]}$$

$$= \frac{\ln (0.0301626752)}{\ln (2.00166436)} = \frac{-3.50115004}{0.69373} = 5.0468466$$



$T = 80^\circ\text{F}$
 $P = 1 \text{ atm}$

$\frac{L}{V} = 1.2$

$L = 1.2V$

$y = mx$

find m

has 4 equilibrium stages

$L \cdot x_0 + V \cdot 1 = V \cdot 0.27 + L \cdot x_N$

$V = V \cdot 0.27 + 1.2V \cdot x_N$

$0.73V = 1.2V \cdot x_N$

$\frac{0.73V}{1.2V} = x_N = 0.60833 \text{ mol}\%$

$y_{N+1} = 1$

$y_1 = 0.27$

$y = mx$

$x_N^* = \frac{y_{N+1}}{m} = \frac{1}{m}$

$N = 4 = \frac{\ln \left[\left(1 - \frac{L}{mV} \right) \left(\frac{x_0 - x_N^*}{x_N - x_N^*} \right) + \frac{L}{mV} \right]}{\ln \left(\frac{mV}{L} \right)}$

$4 = \frac{\ln \left[\left(1 - \frac{1.2}{m} \right) \left(\frac{0 - \frac{1}{m}}{0.60833 - \frac{1}{m}} \right) + \frac{1.2}{m} \right]}{\ln \left(\frac{m}{1.2} \right)}$

$m = 1.413$ by excel solver

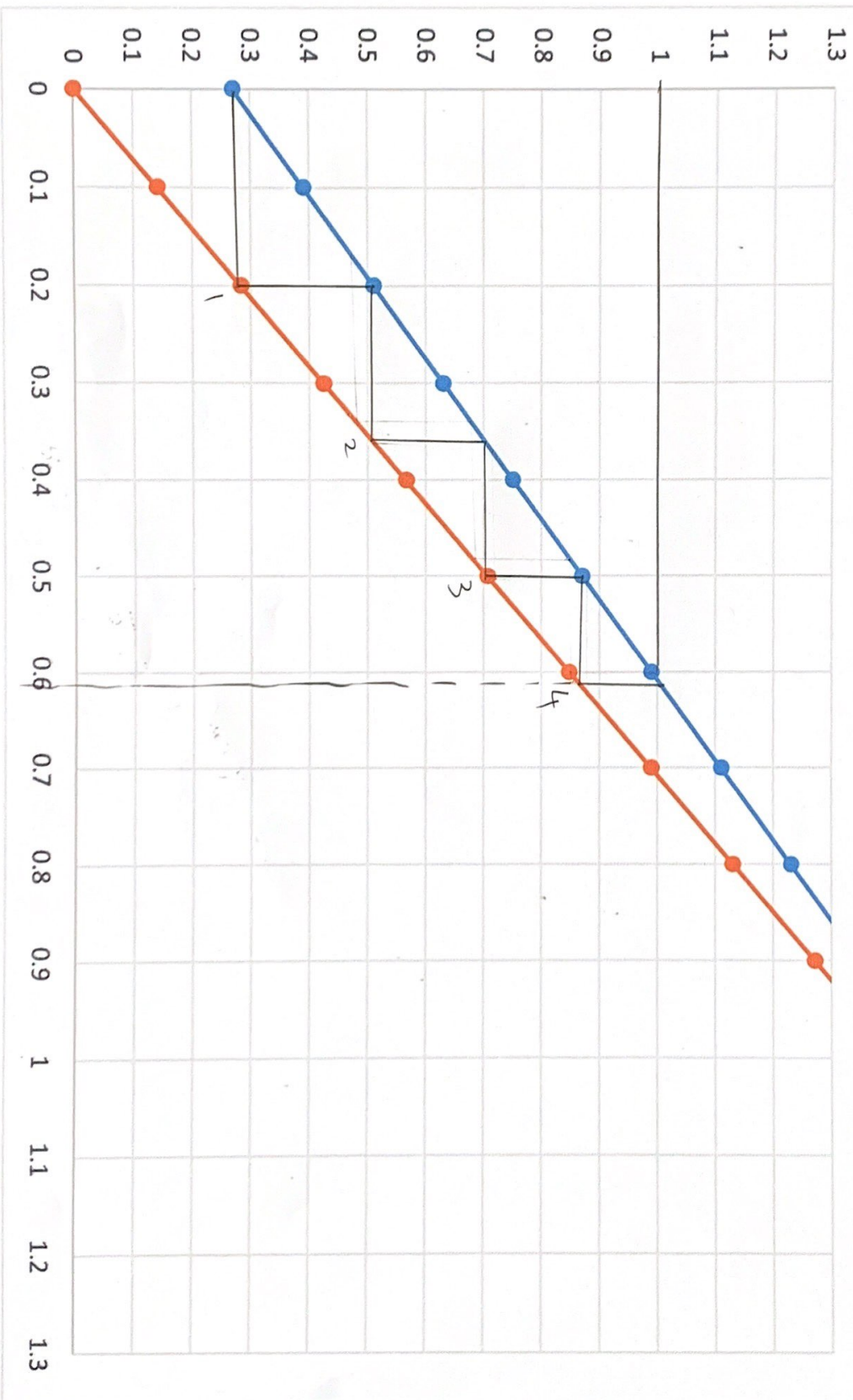
equilibrium: $y = 1.413x$

plot the graph

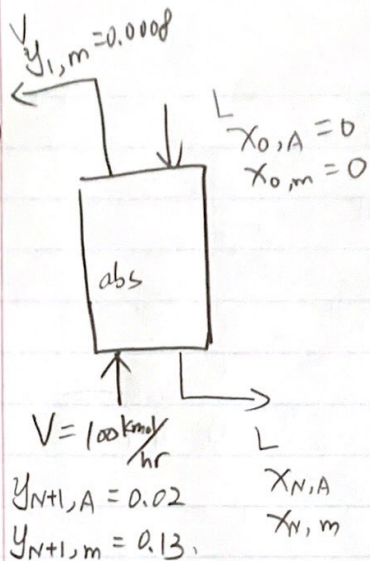
check: by the graph shows exactly 4 stages

so $m = 1.413$ is correct ✓

D11



D16)



L, V are constant

$$\frac{L}{V} = 26$$

$$P_m = 3600 \cdot X_m$$

$$P_A = 7700 \cdot X_A$$

$$H = \frac{P}{P_{\text{tot}}} \quad P_{\text{tot}} = 175 \text{ from D4}$$

$$H_m = \frac{3600}{175} = 20.57143$$

$$H_A = \frac{7700}{175} = 44$$

a) $\frac{L}{V} = 26 \quad L = 26 \cdot V = 2600 \text{ kmol/hr}$

Find $X_{N,m}$

$$L \cdot X_{0,m} + V \cdot y_{N+1,m} = V \cdot y_{1,m} + L \cdot X_{N,m}$$

$$100 \cdot 0.13 = 100 \cdot 0.0008 + 2600 \cdot X_{N,m}$$

$$X_{N,m} = \frac{100 \cdot 0.13 - 100 \cdot 0.0008}{2600} = 0.0049692308 \text{ mol\%}$$

b) $y = Hx$
 $y = 20.57143x$

op line
 $y = \frac{L}{V}x + (y_1 - \frac{L}{V}x_0)$
 $y = 26x + 0.0008$

$$m = 20.57143$$

$$X_N^* = \frac{0.13}{20.57143} = 0.006319444$$

$$N = \frac{\ln \left[\left(1 - \frac{L}{mV} \right) \left(\frac{x_0 - X_N^*}{X_N - X_N^*} \right) + \frac{L}{mV} \right]}{\ln \left(\frac{mV}{L} \right)}$$

$$= \frac{\ln \left[\left(1 - \frac{26}{20.57143} \right) \left(\frac{0 - 0.006319444}{0.0049692308 - 0.006319444} \right) + \frac{26}{20.57143} \right]}{\ln \left(\frac{20.57143}{26} \right)}$$

$$N = \frac{\ln(0.028889212)}{\ln(0.7912088)} = \frac{-3.544287}{-0.2342} = 15.134 \text{ stages}$$

c) Find $y_{1,A}$.

$$\frac{y_{N+1,A} - y_{1,A}}{y_{1,A} - y_{1,A}^*} = \frac{\frac{L}{mV} - \left(\frac{L}{mV}\right)^{N+1}}{1 - \frac{L}{mV}}$$

eq for Ar : $y = 44x$ $m = 44$

$$\begin{aligned} y_{1,A} &= 44 \cdot x_{0,A} \\ &= 44 \cdot 0 = 0 = y_{1,A}^* \end{aligned}$$

$$\frac{0.02 - y_{1,A}}{y_{1,A} - 0} = \frac{\frac{26}{44} - \left(\frac{26}{44}\right)^{15.134+1}}{1 - \frac{26}{44}}$$

$$\frac{0.02 - y_{1,A}}{y_{1,A}} = 1.443941066$$

$$y_{1,A} = 0.008185 \text{ mol\%}$$